Slide Set 8: Double Machine Learning

Debiased ML for Causal Inference in High Dimensions

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Outline

1. The Problem
2. The Partialling-Out Idea
3. The DML Solution
4. Extension: The Interactive Model

Motivation: Treatment Effects with Many Controls

Example: does 401(k) eligibility increase savings?

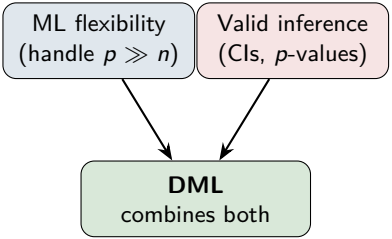
- Y: net financial assets (savings)
- D: 401(k) eligibility (treatment)
- X: income, age, education, family size, marital status, IRA participation, ... ($p = 100+$)

People who are eligible differ systematically from those who are not. We must *adjust for* X.

Three approaches:

1. OLS with all X works if $p \ll n$
2. LASSO / ML to select controls, $O_{\mathbb{P}}(n^{-1/4})$ bias
3. **DML**: valid inference with ML flexibility

The fundamental tension:



This lecture: how DML achieves $\sqrt{n}$-inference on θ_0 while using ML to handle high-dimensional X.

The Partially Linear Model

Model

$$Y_i = \theta_0 D_i + g_0(X_i) + \varepsilon_i$$

- Y_i: outcome
- D_i: treatment (scalar)
- $X_i \in \mathbb{R}^p$: high-dimensional controls
- g_0: unknown nuisance function
- $\mathbb{E}[\varepsilon_i \mid D_i, X_i] = 0$

Goal: estimate θ_0 (causal effect of D on Y, adjusting for X) and form valid confidence intervals.

Key assumptions:

1. **Unconfoundedness:** $\varepsilon_i \perp\!\!\!\perp D_i \mid X_i$
   Conditional on X, treatment is as good as random. No unmeasured confounders.
2. **Partial linearity:** Y depends linearly on D but *nonparametrically* on X via g_0.
3. **High-dimensional:** p may exceed n. Cannot simply "include all controls" in OLS.

Why OLS and Naive ML Both Fail

If $p \ll n$: OLS works. Include all X, get $\hat{\theta}_{OLS} = \theta_0 + O_{\mathbb{P}}(n^{-1/2})$. Done.

If $p \gg n$: OLS is infeasible. Two options:

Option 1: Drop X (short regression)

$$\hat{\theta}_{short} = \theta_0 + \frac{\mathrm{Cov}(D, g_0(X))}{\mathrm{Var}(D)}$$

Confounding bias. Inconsistent.

Option 2: Regularize (LASSO)

$$\min_{\theta,\beta} \sum_i (Y_i - \theta D_i - X_i'\beta)^2 + \lambda \|\beta\|_1$$

Shrinkage creates $O_{\mathbb{P}}(n^{-1/4})$ bias in $\hat{\theta}$.

The bias–variance picture

	Bias	**Variance**
Oracle	0	$O_{\mathbb{P}}(n^{-1/2})$
OLS ($p < n$)	0	$O_{\mathbb{P}}(n^{-1/2})$
LASSO plug-in	$O_{\mathbb{P}}(n^{-1/4})$	$O_{\mathbb{P}}(n^{-1/2})$
DML	$o_{\mathbb{P}}(n^{-1/2})$	$O_{\mathbb{P}}(n^{-1/2})$

For valid 95% CIs, we need bias $\ll$ variance.

LASSO bias is $O_{\mathbb{P}}(n^{-1/4})$, which is *much larger* than the $O_{\mathbb{P}}(n^{-1/2})$ variance.

Goal: bring bias down to $o_{\mathbb{P}}(n^{-1/2})$ so the CLT applies.

Frisch-Waugh-Lovell: Partial Out the Controls

FWL Theorem

In the linear model $\mathbf{Y} = \mathbf{D}\theta + \mathbf{X}\beta + \mathbf{e}$:

$$\hat{\theta}_{OLS} = \frac{\tilde{\mathbf{D}}'\tilde{\mathbf{Y}}}{\tilde{\mathbf{D}}'\tilde{\mathbf{D}}}$$

where $\tilde{\mathbf{D}} = M_X\mathbf{D}$, $\tilde{\mathbf{Y}} = M_X\mathbf{Y}$,
$M_X = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

Recipe:

1. Regress D on $X \Rightarrow$ residual $\tilde{D}$
2. Regress Y on $X \Rightarrow$ residual $\tilde{Y}$
3. Regress $\tilde{Y}$ on $\tilde{D} \Rightarrow$ same $\hat{\theta}$

Only the variation in D *not explained by* X drives the estimate. Confounding variation is removed.

From linear to nonparametric:

In the PLM, g_0 is nonparametric. FWL motivates replacing the linear projection with conditional expectations:

$$m_0(x) := \mathbb{E}[D \mid X{=}x]$$
$$\ell_0(x) := \mathbb{E}[Y \mid X{=}x]$$

These are the nonparametric "best predictions" of D and Y given X. The residuals

$$V_i = D_i - m_0(X_i), \quad \tilde{Y}_i = Y_i - \ell_0(X_i)$$

are the parts not explained by X.

This is Robinson's decomposition (next slide).

Robinson's Decomposition: Eliminating the Nuisance

Take the PLM: $Y_i = \theta_0 D_i + g_0(X_i) + \varepsilon_i$.
Take expectations conditional on X:

$$\ell_0(X_i) = \theta_0\, m_0(X_i) + g_0(X_i)$$

Subtract:

$$Y_i - \ell_0(X_i) = \theta_0(D_i - m_0(X_i)) + \varepsilon_i$$

Robinson's equation

$$\tilde{Y}_i = \theta_0\, V_i + \varepsilon_i$$

The nuisance g_0 has been **eliminated**.

If we knew m_0 and ℓ_0, we could compute V_i and $\tilde{Y}_i$ and run OLS: $\hat{\theta}^{oracle} = \mathbb{E}_n[V_i \tilde{Y}_i]/\mathbb{E}_n[V_i^2]$.

Why the oracle works:

$$\hat{\theta}^{oracle} - \theta_0 = \frac{\mathbb{E}_n[V_i \varepsilon_i]}{\mathbb{E}_n[V_i^2]}$$

Is $\mathbb{E}[V_i \varepsilon_i] = 0$? Yes:

$$\mathbb{E}[V\varepsilon] = \mathbb{E}\big[\underbrace{\mathbb{E}[V \mid X]}_{=0} \cdot \varepsilon\big] = 0$$

because $V = D - \mathbb{E}[D \mid X]$ has mean zero given X.

So $\mathbb{E}_n[V_i \varepsilon_i] = O_{\mathbb{P}}(n^{-1/2})$ by CLT, and the oracle is $\sqrt{n}$-consistent.

The problem: m_0 and ℓ_0 are unknown. We must estimate them with ML.
What goes wrong when we plug in $\hat{m}$ and $\hat{\ell}$?

The Naive Estimator: Why Bias Does Not Vanish

Naive plug-in: replace $g_0 \to \hat{g}$ (estimated with ML).

$$\hat{\theta}^{naive} = \left(\frac{1}{n}\sum D_i^2\right)^{-1} \frac{1}{n}\sum D_i(Y_i - \hat{g}(X_i))$$

Expand the scaled error:

$$\sqrt{n}(\hat{\theta}^{naive} - \theta_0) = \underbrace{a}_{\to N(0,\Sigma)} + \underbrace{b}_{\text{does not vanish}}$$

where:

$$a = \left(\frac{1}{n}\sum D_i^2\right)^{-1} \frac{1}{\sqrt{n}}\sum D_i \varepsilon_i \xrightarrow{d} N(0,\Sigma)$$

$$b = \left(\frac{1}{n}\sum D_i^2\right)^{-1} \frac{1}{\sqrt{n}}\sum D_i(g_0(X_i) - \hat{g}(X_i))$$

Why b does not vanish:

By Cauchy–Schwarz:

$$\left|\frac{1}{n}\sum D_i(g_0 - \hat{g})(X_i)\right| \leq \underbrace{\left\|\frac{D}{\sqrt{n}}\right\|_2}_{O_P(1)} \underbrace{\left\|\frac{g_0 - \hat{g}}{\sqrt{n}}\right\|_2}_{O_P(n^{-1/4})}$$

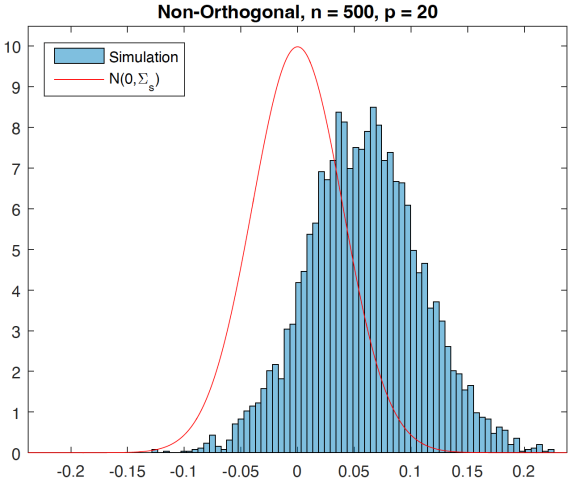
The prefactor $(\frac{1}{n}\sum D_i^2)^{-1} = O_P(1)$, so:

$$b = O_P(1) \cdot \sqrt{n} \cdot O_P(n^{-1/4}) = O_P\left(n^{1/4}\right) \to \infty$$

Bias dominates variance: $b/a = O_P(n^{1/4})$.

$\Rightarrow$ Confidence intervals have coverage $\to 0$.

Bias


Non-Orthogonal, n = 500, p = 20
Simulation
N(0,Σs)
10
9
8
7
6
5
4
3
2
1
0
-0.2
-0.15
-0.1
-0.05
0
0.05
0.1
0.15
0.2

From Naive to DML: The $a + b$ to $a^* + b^* + c^*$ Story

DML estimator uses $\hat{V}_i = D_i - \hat{m}(X_i)$:

$$\hat{\theta}^{DML} = \left(\frac{1}{n}\sum \hat{V}_i D_i\right)^{-1} \frac{1}{n}\sum \hat{V}_i (Y_i - \hat{g}(X_i))$$

Now the scaled error decomposes as:

$$\sqrt{n}(\hat{\theta}^{DML} - \theta_0) = a^* + b^* + c^*$$

where:

$$a^* = (\mathbb{E}[V^2])^{-1} \frac{1}{\sqrt{n}} \sum V_i \varepsilon_i \xrightarrow{d} N(0, \Sigma)$$

$$b^* = (\mathbb{E}[V^2])^{-1} \frac{1}{\sqrt{n}} \sum \underbrace{(\hat{m} - m_0)(X_i)}_{\tilde{m}_i} \underbrace{(\hat{g} - g_0)(X_i)}_{\tilde{\ell}_i}$$

c^* = remaining terms (controlled by cross-fitting)

Why b^* now vanishes:

b^* is a *product* of two ML errors.

$$b^* = O_{\mathbb{P}}(\|\tilde{m}\| \cdot \|\tilde{\ell}\|) = O_{\mathbb{P}}(n^{-1/4} \times n^{-1/4})$$

With Cauchy-Schwarz and the rate condition:

$$b^* = o_{\mathbb{P}}(n^{-1/2}) \checkmark$$

The key structural difference

	Naive	**DML**
Bad term	$D_i \cdot (\hat{g} - g_0)$	$\tilde{m}_i \cdot (\hat{g} - g_0)$
Rate	$O_{\mathbb{P}}(n^{-1/4})$	$O_{\mathbb{P}}(n^{-1/2})$
Vanishes?	No	Yes

Why Cross-Fitting is Essential: Taming c^*

The Problem: Overfitting Bias

If $\hat{g}$ and $\hat{m}$ are trained on the *full* sample, predictions $\hat{g}(X_i)$ are highly correlated with the observation's own noise ε_i.

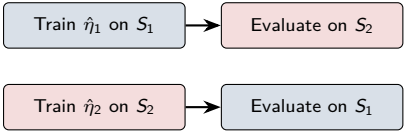
Mathematically, c^* involves empirical process terms:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left(\text{Error}_i(\hat{\eta}) - \mathbb{E}[\text{Error}_i(\hat{\eta})] \right)$$

For this to vanish without sample splitting, the ML hypothesis space must be strictly limited in its complexity (it must be a **Donsker class**).

The Fatal Flaw: Modern ML models are too complex and *fail* the Donsker condition.

The Solution: Cross-Fitting



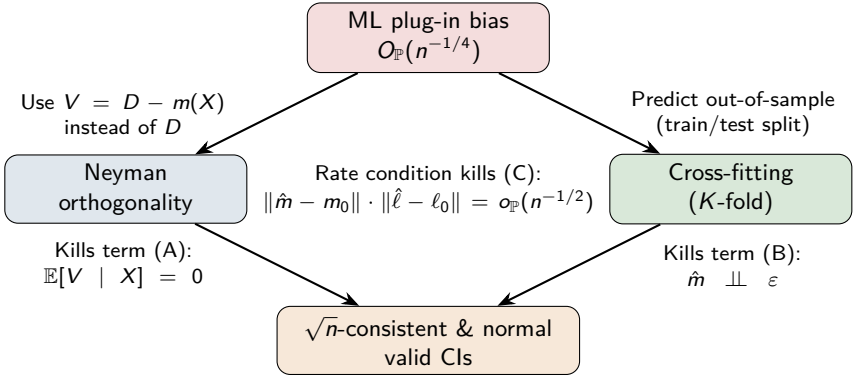
By evaluating on hold-out folds, the nuisance estimator $\hat{\eta}_{-k}$ is strictly **independent** of observation i.

Because $\hat{\eta}_{-k}$ is predetermined relative to fold k, the expected value of the cross-terms perfectly zeroes out:

$$\mathbb{E}\left[V_i \left(\hat{g}_{-k}(X_i) - g_0(X_i) \right) \,\middle|\, S_{-k} \right] = 0$$

(This holds because $\mathbb{E}[V_i \mid X_i] = 0$).

The Big Picture: How DML Fixes the Bias



Ingredient 1: Neyman Orthogonality

The naive score uses D_i directly:

$$\psi^{naive} = (Y_i - D_i\theta - \hat{g}(X_i)) \cdot D_i$$

If $\hat{g}$ has errors, they feed directly into $\hat{\theta}$.

The DML score uses $V_i = D_i - m(X_i)$:

$$\psi^{DML} = (\tilde{Y}_i - \theta V_i) \cdot V_i$$

Why does this help?
V_i is "centered": $\mathbb{E}[V \mid X] = 0$. So when $\hat{\ell}$ has errors $\tilde{\ell}(X)$, their effect is:

$$\mathbb{E}[\tilde{\ell}(X) \cdot V] = \mathbb{E}[\tilde{\ell}(X) \cdot \underbrace{\mathbb{E}[V \mid X]}_{=0}] = 0$$

The errors are orthogonal to the score.

Neyman orthogonality

A score $\psi(W; \theta, \eta)$ is **Neyman orthogonal** if perturbing the nuisance η around the truth does not change the expected score:

$$\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta_0)][\delta\eta] = 0$$

for all directions $\delta\eta$.

Consequence:
Errors in $\hat{\eta}$ only affect $\hat{\theta}$ at *second order*:

$$\text{bias} = O_{\mathbb{P}}(\|\hat{\eta} - \eta_0\|^2) \quad \text{instead of} \quad O_{\mathbb{P}}(\|\hat{\eta} - \eta_0\|)$$

This converts an $n^{-1/4}$ problem into an $n^{-1/2}$ problem.

Checking Orthogonality for the PLM Score

The DML score written in full:

$$\psi = \big(Y - D\theta - \ell(X) + \theta\, m(X)\big)\big(D - m(X)\big)$$

At the truth: $\psi = \varepsilon_i \cdot V_i$.

<u>Perturb ℓ</u> (the outcome nuisance):

$$\begin{aligned}\partial_\ell \mathbb{E}[\psi][\delta\ell] &= -\mathbb{E}[\delta\ell(X) \cdot V] \\ &= -\mathbb{E}\big[\delta\ell(X) \cdot \underbrace{\mathbb{E}[V \mid X]}_{=0}\big] \\ &= 0 \checkmark\end{aligned}$$

Why: $V = D - m_0(X)$ has mean zero given X by construction.

<u>Perturb m</u> (the propensity nuisance):

$$\begin{aligned}\partial_m \mathbb{E}[\psi][\delta m] &= -\mathbb{E}[\varepsilon \cdot \delta m(X)] \\ &= -\mathbb{E}\big[\delta m(X) \cdot \underbrace{\mathbb{E}[\varepsilon \mid X]}_{=0}\big] \\ &= 0 \checkmark\end{aligned}$$

Why: $\varepsilon = Y - \theta_0 D - g_0(X)$ has mean zero given X by assumption.

Two orthogonalities, two reasons

- Errors in $\hat{\ell}$ washed out because $\mathbb{E}[V \mid X] = 0$
- Errors in $\hat{m}$ washed out because $\mathbb{E}[\varepsilon \mid X] = 0$

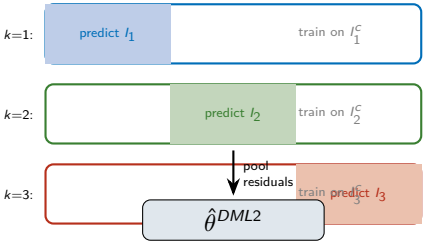
Ingredient 2: Cross-Fitting

Problem: even with an orthogonal score, if $\hat{m}$ is trained on the *same data* used to evaluate $\hat{V}_i \varepsilon_i$, overfitting makes them correlated.
This is term (B): $\mathbb{E}_n[\tilde{m}_i \varepsilon_i] \neq 0$.

Fix: predict out-of-sample.

K-fold cross-fitting

1. Split data into K folds $I_1, \ldots, I_K$
2. For fold k:
   - Train $\hat{m}_{-k}$, $\hat{\ell}_{-k}$ on I_k^c
   - Predict on I_k:
     $\hat{V}_i = D_i - \hat{m}_{-k}(X_i)$
     $\hat{\tilde{Y}}_i = Y_i - \hat{\ell}_{-k}(X_i)$
3. Pool all residuals across folds



Key: $\hat{m}_{-k}(X_i)$ was trained *without* seeing observation i, so it is independent of ε_i.
This kills term (B).

Are the Rate Conditions Actually Achievable?

DML requires each nuisance estimator to achieve $\|\hat{\eta} - \eta_0\|_n = o_{\mathbb{P}}(n^{-1/4})$.

This is exactly what we studied in earlier lectures.

Method	**Rate**	**Condition**
LASSO	$\sqrt{\frac{s \log p}{n}}$	$s \log p = o_{\mathbb{P}}(\sqrt{n})$
Random forest	$n^{-\frac{\beta}{2\beta+d}}$	$\beta > d/2$
Neural net	near minimax	sparsity / low dim

Each rate must be $o(n^{-1/4})$ for the product condition to hold.

LASSO (Lecture 2): under s-sparsity,

$$\|\hat{\beta} - \beta_0\|_2 = O_{\mathbb{P}}\left(\sqrt{\frac{s \log p}{n}}\right)$$

For $n^{-1/4}$: need $s \log p = o_{\mathbb{P}}(\sqrt{n})$.
With $p = 100$ controls and $s = 5$ truly nonzero: $s \log p \approx 23 \ll \sqrt{n}$ even for $n = 500$. **Valid.**

Random forest (Lecture 3): under β-smoothness in d dimensions,

$$\|\hat{f} - f_0\|_2 = O_{\mathbb{P}}(n^{-\beta/(2\beta+d)})$$

For $n^{-1/4}$: need $\beta > d/2$. With $d = 5$, need $\beta > 2.5$: moderate smoothness. **Reasonable.**
Neural networks (Lecture 4): can adapt to low effective dimension. Under sparsity or smoothness, near-minimax rates achievable.

Beyond Partial Linearity: Estimating the ATE

What if the treatment effect varies with X?

Binary treatment $D \in \{0, 1\}$, potential outcomes Y^0, Y^1.

Interactive model:

$$Y_i = \ell_0(D_i, X_i) + \varepsilon_i$$

where $\ell_0(d, x) = \mathbb{E}[Y \mid D{=}d, X{=}x]$ is fully nonparametric. No linearity in D.

Estimand:

$$\theta_0 = \mathrm{ATE} = \mathbb{E}[Y^1 - Y^0] = \mathbb{E}[\ell_0(1, X) - \ell_0(0, X)]$$

Assumptions:

- Unconfoundedness: $Y^d \perp\!\!\!\perp D \mid X$
- Overlap: $0 < P(D{=}1 \mid X) < 1$

The same bias problem applies.

Plug-in: $\hat{\theta} = \mathbb{E}_n[\hat{\ell}(1, X_i) - \hat{\ell}(0, X_i)]$
Bias: $O_{\mathbb{P}}(\|\hat{\ell} - \ell_0\|) = O_{\mathbb{P}}(n^{-1/4})$.

IPW: $\hat{\theta} = \mathbb{E}_n\left[\frac{D_i Y_i}{\hat{m}(X_i)} - \frac{(1-D_i)Y_i}{1-\hat{m}(X_i)}\right]$
Bias: $O_{\mathbb{P}}(\|\hat{m} - m_0\|) = O_{\mathbb{P}}(n^{-1/4})$.

Doubly robust (DML): Combines plug-in + IPW correction.
Bias: $O_{\mathbb{P}}(\|\hat{\ell} - \ell_0\| \cdot \|\hat{m} - m_0\|) = O_{\mathbb{P}}(n^{-1/2})$.

Key insight

DR combines two imperfect estimators. Neither alone achieves $n^{-1/2}$, but together (via Neyman orthogonality) they do.

The Doubly Robust Score

DR score for ATE

$$\psi_{DR} = \underbrace{\ell(1,X) - \ell(0,X)}_{\text{plug-in}} - \theta + \underbrace{\frac{D(Y-\ell(1,X))}{m(X)} - \frac{(1-D)(Y-\ell(0,X))}{1-m(X)}}_{\text{IPW bias correction}}$$

Structure: start with the plug-in estimate, then correct its bias using inverse propensity weighting on the residuals $Y - \hat{\ell}(d, X)$.

Doubly robust: $\mathbb{E}[\psi_{DR}] = 0$ if *either* $\ell = \ell_0$ or $m = m_0$. Consistent if at least one model is correct.

Neyman orthogonal: perturbations in ℓ and m around the truth both vanish (same logic as PLM).

DML-ATE estimator (with cross-fitting):

$$\hat{\theta}^{DML}_{ATE} = \mathbb{E}_n\left[\hat{\ell}(1,X_i) - \hat{\ell}(0,X_i) + \frac{D_i\hat{\varepsilon}^1_i}{\hat{m}(X_i)} - \frac{(1-D_i)\hat{\varepsilon}^0_i}{1-\hat{m}(X_i)}\right]$$

where $\hat{\varepsilon}^d_i = Y_i - \hat{\ell}(d, X_i)$.

Result:

$$\sqrt{n}(\hat{\theta}^{DML}_{ATE} - \theta_0) \xrightarrow{d} N\big(0,\ \mathrm{Var}(\psi_{DR})\big)$$

This achieves the **semiparametric efficiency bound** (Robins & Rotnitzky 1995, Hahn 1998).

Verifying Orthogonality for the DR Score

Perturb the outcome model ℓ:

The treated-arm contribution:

$$\mathbb{E}\left[\delta\ell(1,X)\left(1-\frac{D}{m_0(X)}\right)\right]$$

$$= \mathbb{E}\left[\delta\ell(1,X)\left(1-\frac{\overbrace{\mathbb{E}[D \mid X]}^{=m_0(X)}}{m_0(X)}\right)\right]$$

$$= \mathbb{E}[\delta\ell(1,X)\cdot 0] = 0 \checkmark$$

Why: the propensity score $m_0(X) = \mathbb{E}[D \mid X]$ exactly cancels the IPW denominator. Similarly for $\delta\ell(0,X)$. ✓

Perturb the propensity m:

Need: $\mathbb{E}\big[D(Y-\ell_0(1,X)) \mid X\big] = 0.$

$$\mathbb{E}\big[D(Y-\ell_0(1,X)) \mid X\big]$$

$$= m_0(X)\,\underbrace{\mathbb{E}[Y-\ell_0(1,X) \mid D{=}1, X]}_{=0}$$

$$= 0 \checkmark$$

Why: $\ell_0(1,x) = \mathbb{E}[Y \mid D{=}1, X{=}x]$.

Both directions orthogonal

- Errors in $\hat{\ell}$: canceled by propensity weighting and in $\hat{m}$: canceled by outcome residuals